

Perfectoid Moduli of Quotient Bundles on the Fargues–Fontaine Curve

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Abstract

Let $F/\mathbf{Q}_p$ be a finite extension and let C be a complete algebraically closed perfectoid extension of F . For $1 \leq d < n$, let $W_{n,d}$ be the moduli v -sheaf of quotient bundles

$$\mathcal{O}_{X_S}(1/n) \twoheadrightarrow Q$$

on relative Fargues–Fontaine curves whose geometric fibres have type $\mathcal{O}(1/d)$. Hansen–Mann showed that $W_{n,d}$ is a spatial diamond. We prove that it is represented by a perfectoid space. This proves their Conjecture 5.4 and, together with their cohomological results, gives the expected upper vanishing bound for the derived p -adic Jacquet–Langlands functor on parabolically induced representations.

1 Introduction

Moduli of vector bundles on the Fargues–Fontaine curve provide a geometric setting for the p -adic Jacquet–Langlands correspondence. Problem 1.1 in Hansen’s problem list [7], formulated precisely as Conjecture 5.4 of Hansen–Mann [8], asks whether certain quotient-bundle moduli diamonds are represented by perfectoid spaces. We prove this conjecture.

The relevant object $W_{n,d}$ parametrizes quotients of the stable rank- n , degree-one bundle whose geometric fibres are stable of rank d and degree one. Hansen–Mann already show that $W_{n,d}$ is a spatial diamond [8, Definition 5.1 and the following discussion]; the issue is whether this diamond comes from a perfectoid space. This stronger statement is useful for two reasons. It gives genuine affinoid perfectoid charts, hence access to the analytic structure sheaf, pro-étale methods and tilting on $W_{n,d}$ itself; and it is the geometric input needed to combine Hansen–Mann’s description of the derived Jacquet–Langlands functor with their cohomological dimension estimate. Thus the representability result has a direct representation-theoretic consequence, recorded below.

Throughout, $F/\mathbf{Q}_p$ is a fixed finite extension and C is a fixed complete algebraically closed perfectoid extension of F . We write Perf_C for perfectoid spaces over C with the v -topology. For $S \in \mathrm{Perf}_C$, let X_S be the relative Fargues–Fontaine curve attached to $S^\flat$ and F . We write C° for the ring of power-bounded elements of C and put

$$X := X_{\mathrm{Spa}(C, C^\circ)}.$$

For $\lambda \in \mathbf{Q}$, we write $\mathcal{O}_{X_S}(\lambda)$ for the standard vector bundle of slope λ . We use the theory of v -sheaves and diamonds in the sense of Scholze [4].

Convention 1.1 (Global and geometric bundle type). For vector bundles E, E' on X_S , the notation $E \cong_{X_S} E'$ means that a global isomorphism exists on X_S , while

$$E \cong_{\mathrm{geom}/S} \mathcal{O}(\lambda)^{\oplus m}$$

means that for every geometric point $\bar{s} \rightarrow S$ there exists an isomorphism

$$E|_{X_{\bar{s}}} \cong_{X_{\bar{s}}} \mathcal{O}_{X_{\bar{s}}}(\lambda)^{\oplus m}.$$

For bundles on X we write $\cong_X$, and we reserve $\simeq$ for equivalences of functors, v -sheaves, diamonds, spaces, or categories. Unless explicitly named, these symbols assert existence only and do not include a choice; choices used in the proof, such as α_t in Section 3, are named explicitly.

We fix integers $1 \leq d < n$ and set

$$V := \mathcal{O}_X(1/n), \quad r := n - d.$$

For $S \in \mathrm{Perf}_C$, let $\mathcal{Q}_{n,d}(S)$ be the groupoid of surjections

$$q : V_S \twoheadrightarrow Q, \quad Q \cong_{\mathrm{geom}/S} \mathcal{O}(1/d),$$

with morphisms the isomorphisms of quotients commuting with q . Since q is surjective, every automorphism is trivial; equivalently, this groupoid is 0-truncated. We define

$$W_{n,d} := a_v(S \mapsto \pi_0 \mathcal{Q}_{n,d}(S)), \quad (1.1)$$

where a_v denotes v -sheafification. Thus no sheafification is tacit in the notation. This is the v -sheaf of [8, Definition 5.1], hence a spatial diamond by the discussion following that definition.

Theorem 1.2. *Let $F/\mathbf{Q}_p$ be a finite extension and let C be a complete algebraically closed perfectoid extension of F . For every $1 \leq d < n$, the v -sheaf $W_{n,d}$ over C is represented by a perfectoid space over C .*

Besides providing genuine perfectoid charts on the known spatial diamond $W_{n,d}$, Theorem 1.2 gives the hypothesis used in Hansen–Mann’s cohomological estimate. Their Theorem 5.2 describes the derived Jacquet–Langlands functor of a parabolically induced representation by cohomology on $W_{n,d}$; together with [8, Proposition 5.5] we obtain

$$\mathcal{J}_n^i \left(\mathrm{Ind}_{P_{n-d,d}(F)}^{\mathrm{GL}_n(F)} (\sigma_1 \boxtimes \sigma_2) \right) = 0 \quad (i > n + d - 2), \quad (1.2)$$

where $P_{n-d,d}$ has Levi $\mathrm{GL}_{n-d} \times \mathrm{GL}_d$, the σ_i are admissible smooth $\mathbf{F}_p$ -representations of the two Levi factors, and $\mathcal{J}_n^i$ denotes the i th cohomology of the derived Jacquet–Langlands functor. For $d = 1$, $W_{n,1}$ is the perfectoid parabolic quotient of Johansson–Ludwig–Hansen [3, Theorem A, Proposition 4.2.7 and Theorem A.1.1]; for $d > 1$, it should not be confused with the full parabolic quotient of block type $(n-d, d)$, which is not perfectoid [3, Theorem A.1.5].

Strategy of the proof. We argue by induction on d , starting from $d = 1$. Writing $r = n - d$, an object of $W_{n,d}$ is equivalently a kernel $K \hookrightarrow V_S$ of geometric type $\mathcal{O}^{\oplus r}$. For a fixed global section $t : \mathcal{O}_X \hookrightarrow V$, let U_t be the open locus on which $K \oplus \mathcal{O}_{t,S} \rightarrow V_S$ is stably injective. Enlarging K by $\mathcal{O}_{t,S}$ and then quotienting by this line gives

$$U_t \longrightarrow \mathcal{V}_t \simeq W_{n-1,d-1}.$$

This map is the torsor of splittings of an extension of slope-zero bundles, under a finite-rank F -local system, so U_t is perfectoid by induction. Finally, a fixed trivial rank- $(r+1)$ subbundle of V supplies finitely many sections whose transverse loci cover every point. The fibrewise-injectivity formalism of Anschütz–Le Bras [6, §2.3] and their openness argument [6, Remark 2.15, footnote 5] are used throughout.

2 Formal reductions

2.1 Open-local representability

Lemma 2.1 (Open-local representability). *Suppose that Y is a diamond and that every point $y \in |Y|$ is contained in an open subdiamond $U_y \subset Y$ represented by a perfectoid space. Then Y is represented by a perfectoid space.*

Proof. By Scholze’s correspondence between open subdiamonds and open subsets [4, Proposition 11.15], the U_y form an open covering. Their intersections are open perfectoid subspaces, so the representing perfectoid spaces glue to a perfectoid space representing Y . $\square$

2.2 Kernels and the diamond structure

Lemma 2.2 (Kernel formulation). *Suppose that $F/\mathbf{Q}_p$ and C are as in Theorem 1.2, and suppose that S is a perfectoid space over C . With $V = \mathcal{O}_X(1/n)$ and $r = n - d$, the v -sheaf $W_{n,d}$ is naturally equivalent to the v -sheaf of isomorphism classes of stably injective maps*

$$K \longrightarrow V_S, \quad K \cong_{\text{geom}/S} \mathcal{O}^{\oplus r}. \quad (2.1)$$

Here “stably injective” means injective after every base change $S' \rightarrow S$. Equivalently, it may be checked after pullback to every geometric point of S . This is the fibrewise injectivity condition used in [6, §2.3, before Definition 2.7]; see also the discussion preceding [3, Lemma A.2.2]. Isomorphisms between two kernels in (2.1) are required to commute with their maps to V_S .

Proof. We first pass from quotients to kernels. Suppose that

$$q : V_S \rightarrow Q$$

represents an S -point of $W_{n,d}$ before v -sheafification, and set $K = \ker(q)$. We explain why K is a vector bundle and why its inclusion is stably injective.

On a sufficiently small Tate affinoid open $U = \text{Spa}(R, R^+) \subset X_S$, vector bundles correspond to finite projective R -modules by [5, Theorem 2.7.7]. After shrinking U , local basis sections of Q lift through q , so the restriction of q is represented by a surjection

$$M \twoheadrightarrow N$$

of finite projective R -modules. Because N is projective, the surjection splits. Its kernel is therefore finite projective, and the resulting short exact sequence remains exact after arbitrary base change. These local kernels glue to a vector bundle K , and $K \hookrightarrow V_S$ remains injective after every base change. We will repeatedly use this same local splitting argument below whenever an injection has vector-bundle cokernel.

We next determine the geometric type of K . For a geometric point $\bar{s} \rightarrow S$, the bundle $K_{\bar{s}}$ has rank r and degree 0. On the absolute Fargues–Fontaine curve, the Harder–Narasimhan filtration splits, and the stable bundle of reduced positive slope a/b has rank b [2, Theorem I.3.2]. Suppose that $K_{\bar{s}}$ had a positive stable summand $\mathcal{O}(a/b)$. Then $b \leq r < n$, and hence

$$\frac{a}{b} \geq \frac{1}{b} > \frac{1}{n}.$$

The restriction of the injection $K_{\bar{s}} \hookrightarrow \mathcal{O}(1/n)$ to this summand would give a nonzero section of

$$\text{Hom}(\mathcal{O}(a/b), \mathcal{O}(1/n)),$$

a semistable bundle of slope $1/n - a/b < 0$. This contradicts [2, Proposition II.2.5(i)]. Therefore every slope of $K_{\bar{s}}$ is nonpositive. Since its total degree is zero, all slopes are zero, and the classification of bundles gives

$$K_{\bar{s}} \cong \mathcal{O}^{\oplus r}.$$

Thus $K \cong_{\text{geom}/S} \mathcal{O}^{\oplus r}$.

Conversely, suppose that $K \rightarrow V_S$ is stably injective and that $K \cong_{\text{geom}/S} \mathcal{O}^{\oplus r}$. Lemma A.2.2 of [3] shows that V_S/K is a vector bundle. Fibrewise its rank is d and its degree is 1, and the same lemma identifies its geometric bundle type as

$$V_S/K \cong_{\text{geom}/S} \mathcal{O}\left(\frac{1}{n-r}\right) = \mathcal{O}(1/d).$$

Hence $V_S \twoheadrightarrow V_S/K$ defines a point of $W_{n,d}$.

The two constructions are inverse and commute with base change. Since both descriptions are compatible with v -descent, they induce the asserted equivalence of v -sheaves. $\square$

Proposition 2.3. *The v -sheaf $W_{n,d}$ is a spatial diamond, and hence in particular a diamond.*

Proof. This is the quotient-bundle v -sheaf of [8, Definition 5.1], which is shown in the discussion following that definition to be a spatial diamond. $\square$

2.3 The transverse open locus

We now fix a stably injective map on the absolute curve

$$t : \mathcal{O}_X \hookrightarrow V.$$

If $K \hookrightarrow V_S$ is an S -point of $W_{n,d}$ in the kernel description, we write

$$(K, t)_S : K \oplus \mathcal{O}_{t,S} \longrightarrow V_S$$

for the sum of the two inclusions. We define the subfunctor $U_t \subset W_{n,d}$ by

$$U_t(S) := \{K \hookrightarrow V_S \in W_{n,d}(S) \mid (K, t)_S \text{ is stably injective}\}. \quad (2.2)$$

Lemma 2.4 (Openness of the transverse locus). *The inclusion $U_t \hookrightarrow W_{n,d}$ is an open immersion of diamonds.*

Proof. Let $S \rightarrow W_{n,d}$ be a perfectoid test space, with quotient $q : V_S \rightarrow Q$ and kernel K . The composite

$$\bar{t} = q \circ t_S : \mathcal{O}_{X_S} \longrightarrow Q$$

defines a section of the relative Banach–Colmez space $\mathcal{B}_Q(T) = H^0(X_T, Q_T)$. By [2, Proposition II.3.5(i)], applied to $[0 \rightarrow Q]$, this is a locally spatial diamond, partially proper and hence separated over S ; its zero section is therefore closed. Let $S^\circ \subset S$ be the inverse image of its complement.

At a geometric point $\bar{s} \rightarrow S$, the curve $X_{\bar{s}}$ is integral, so $\bar{t}_{\bar{s}}$ is nonzero if and only if it is injective, equivalently if $K_{\bar{s}} \oplus \mathcal{O}_{t,\bar{s}} \rightarrow V_{\bar{s}}$ is injective. By Lemma 2.2, this is precisely the fibrewise criterion for stable injectivity. Hence $S^\circ = U_t \times_{W_{n,d}} S$, proving that $U_t \hookrightarrow W_{n,d}$ is open. This is the rank-one form of [6, Remark 2.15, footnote 5]. $\square$

3 The two-step reduction attached to a section

Throughout this section $d > 1$, and $t : \mathcal{O}_X \hookrightarrow V$ is stably injective. By [3, Lemma A.2.2],

$$V^t := V/\mathcal{O}_t$$

is a vector bundle of rank $n - 1$, degree 1, and geometric type $\mathcal{O}(1/(n - 1))$. The classification [2, Theorem I.3.2] therefore gives a global isomorphism

$$\alpha_t : V^t \xrightarrow{\sim}_X \mathcal{O}_X(1/(n - 1)), \quad (3.1)$$

which we choose once and for all.

3.1 The incidence diamond inside $W_{n,d-1}$

We define $\mathcal{V}_t \subset W_{n,d-1}$ by requiring that t_S factor through the kernel $\widetilde{K}$ and that $\widetilde{K}/\mathcal{O}_{t,S}$ be a vector bundle. Then $\mathcal{O}_{t,S} \hookrightarrow \widetilde{K}$ is locally split and hence remains injective after base change.

Proposition 3.1 (Incidence reduction). *There is a natural equivalence*

$$\mathcal{V}_t \simeq W_{n-1,d-1}.$$

After the choice of α_t in (3.1), the equivalence sends

$$\widetilde{K} \hookrightarrow V_S \quad \longmapsto \quad J := \widetilde{K}/\mathcal{O}_{t,S} \hookrightarrow V_S^t \xrightarrow{\alpha_t} \mathcal{O}_{X_S}(1/(n - 1)).$$

Proof. For $\widetilde{K} \in \mathcal{V}_t(S)$, the bundle $J = \widetilde{K}/\mathcal{O}_{t,S}$ satisfies

$$V_S^t/J \cong V_S/\widetilde{K} \cong_{\text{geom}/S} \mathcal{O}(1/(d - 1)),$$

so it defines a point of $W_{n-1,d-1}$; Lemma 2.2 identifies J as a kernel of geometric type $\mathcal{O}^{\oplus r}$.

Conversely, for a quotient $V_S^t \twoheadrightarrow Q$ in $W_{n-1,d-1}$ with kernel J , we set

$$\widetilde{K} := V_S \times_{V_S^t} J = \ker(V_S \twoheadrightarrow V_S^t \twoheadrightarrow Q).$$

Since Q is a vector bundle, Lemma 2.2's local splitting argument shows that $\widetilde{K}$ is a vector bundle, stably injective in V_S , and fits into

$$0 \longrightarrow \mathcal{O}_{t,S} \longrightarrow \widetilde{K} \longrightarrow J \longrightarrow 0. \quad (3.2)$$

Moreover $V_S/\widetilde{K} \cong Q$, so $\widetilde{K} \in \mathcal{V}_t(S)$. The constructions are inverse and commute with base change. $\square$

3.2 The reduction morphism

For $K \hookrightarrow V_S$ in $U_t(S)$, the stably injective sum has image

$$\widetilde{K}_t(K) := K + \mathcal{O}_{t,S} \cong_{X_S} K \oplus \mathcal{O}_{t,S}.$$

By [3, Lemma A.2.2],

$$V_S/\widetilde{K}_t(K) \cong_{\text{geom}/S} \mathcal{O}\left(\frac{1}{n - (r + 1)}\right) = \mathcal{O}(1/(d - 1)).$$

Hence

$$\sigma^{(t)} : U_t \longrightarrow W_{n,d-1}, \quad K \longmapsto K + \mathcal{O}_t \quad (3.3)$$

has image in $\mathcal{V}_t$, and factors as

$$\rho_t : U_t \longrightarrow \mathcal{V}_t \simeq W_{n-1,d-1}. \quad (3.4)$$

3.3 The splitting torsor

Assume that $W_{n-1,d-1}$ is represented by a perfectoid space; then so is $\mathcal{V}_t$. Let $J_t^{\text{univ}} \hookrightarrow V_{\mathcal{V}_t}^t$ be the universal kernel and $\widetilde{K}_t^{\text{univ}}$ its inverse image in $V_{\mathcal{V}_t}$. We obtain

$$e_t : 0 \longrightarrow \mathcal{O}_{t,\mathcal{V}_t} \longrightarrow \widetilde{K}_t^{\text{univ}} \longrightarrow J_t^{\text{univ}} \longrightarrow 0. \quad (3.5)$$

We write $\text{Spl}_{\mathcal{V}_t}(e_t)$ for the v -sheaf of splittings of this extension.

Proposition 3.2 (The chart is a splitting torsor). *The morphism $\rho_t : U_t \rightarrow \mathcal{V}_t$ induces a natural equivalence over $\mathcal{V}_t$*

$$U_t \simeq \text{Spl}_{\mathcal{V}_t}(e_t).$$

The splitting sheaf is a torsor under the relative Hom sheaf

$$\mathcal{H}_t(S) = \text{Hom}_{X_S}(J_{t,S}^{\text{univ}}, \mathcal{O}_{t,S}), \quad S \longrightarrow \mathcal{V}_t, \quad (3.6)$$

and $\mathcal{H}_t$ is a pro-étale F -local system of rank r .

Proof. For $K \in U_t(S)$, the decomposition

$$\widetilde{K}_t(K) = K \oplus_{X_S} \mathcal{O}_{t,S}$$

is a splitting of the pullback of (3.5). Conversely, a splitting $s : J \rightarrow \widetilde{K}$ has image K with $\widetilde{K} = \mathcal{O}_{t,S} \oplus K$. Thus $K \cong J$ is a vector bundle of geometric type $\mathcal{O}^{\oplus r}$; the local splitting argument of Lemma 2.2 shows that $K \hookrightarrow V_S$ and $K \oplus \mathcal{O}_{t,S} \rightarrow V_S$ are stably injective, and [3, Lemma A.2.2] gives $V_S/K \cong_{\text{geom}/S} \mathcal{O}(1/d)$. These constructions are inverse.

The difference of two splittings is a unique map from the quotient to the kernel, so the splitting sheaf is a torsor under $\mathcal{H}_t$. The extension class lies in

$$\text{Ext}_{X_{\mathcal{V}_t}}^1(J_t^{\text{univ}}, \mathcal{O}_{t,\mathcal{V}_t}) = H^1(X_{\mathcal{V}_t}, (J_t^{\text{univ}})^{\vee}).$$

The dual bundle has geometric slope zero, so [2, Proposition II.3.4(ii)] kills this class after a pro-étale cover; splittings therefore exist pro-étale locally. Finally, [2, Corollary II.2.20] identifies $\mathcal{H}_t$ with the dual of the rank- r F -local system corresponding to J_t^{univ} . $\square$

Lemma 3.3 (Torsors under an F -local system). *Suppose that B is a perfectoid space, L is a finite-rank pro-étale F -local system on B , and $P \rightarrow B$ is a pro-étale torsor under the additive group L . Then P is represented by a perfectoid space, and the structural map $P \rightarrow B$ is pro-étale.*

Proof. The assertion is local for open coverings of B . We first regard L as a $\mathbf{Q}_p$ -local system. By [5, Proposition 8.4.6 and Lemma 9.1.11], after passing to an affinoid open covering of B we may choose a $\mathbf{Z}_p$ -local system $\Lambda \subset L$ with $L = \Lambda[1/p]$. We work over one such affinoid open and write $N = \text{rk}_{\mathbf{Z}_p} \Lambda$.

For every $k \geq 0$, we form the quotient v -sheaf

$$P_k := P/(p^k \Lambda).$$

There exists a common pro-étale cover of the base on which both Λ and the torsor P become trivial. On this cover, P_0 is a disjoint union of copies of the base indexed by $\mathbf{Q}_p^N/\mathbf{Z}_p^N$. The separated étale descent theorem [4, Proposition 9.7], which applies over an arbitrary perfectoid base, therefore represents P_0 by a perfectoid space separated and étale over B .

On the same trivializing cover, the transition map

$$P_{k+1} \longrightarrow P_k$$

has fibre

$$p^k \mathbf{Z}_p^N / p^{k+1} \mathbf{Z}_p^N.$$

It is consequently finite étale, and finite étale descent in [4, Proposition 9.7] shows that every P_k is represented by a perfectoid space and that all transition maps are finite étale.

There is a natural map of v -sheaves

$$P \longrightarrow \varprojlim_{k \geq 0} P_k.$$

It is an isomorphism. This can be checked after the common trivializing pro-étale cover, where it becomes the standard isomorphism of topological groups

$$\mathbf{Q}_p^N \xrightarrow{\sim} \varprojlim_k \mathbf{Q}_p^N / p^k \mathbf{Z}_p^N.$$

Completeness gives surjectivity and $\bigcap_k p^k \mathbf{Z}_p^N = 0$ gives injectivity.

Now suppose that $U \subset P_0$ is an affinoid perfectoid open. Each base change $P_k \times_{P_0} U$ is affinoid and finite étale over U . By [4, Proposition 6.5 and Definition 7.8], the inverse limit of this tower is an affinoid perfectoid space, pro-étale over U . These inverse limits represent the open restrictions $P \times_{P_0} U$ and glue as U varies. Thus P is represented by a perfectoid space, pro-étale over P_0 , and therefore pro-étale over B . The constructions on the original affinoid open covering of B represent restrictions of the same v -sheaf P , so they glue globally over B . $\square$

Corollary 3.4. *If $W_{n-1,d-1}$ is represented by a perfectoid space, then U_t is represented by a perfectoid space.*

Proof. By Propositions 3.1 and 3.2, U_t is a torsor over the perfectoid space $\mathcal{V}_t \simeq W_{n-1,d-1}$ under the F -local system $\mathcal{H}_t$. We apply Lemma 3.3. $\square$

4 Transverse global sections

Lemma 4.1 (A fixed supply of global transverse sections). *Suppose that $d > 1$. There is a short exact sequence on the absolute Fargues–Fontaine curve X*

$$0 \longrightarrow A \longrightarrow V \longrightarrow \mathcal{O}_X(1/(d-1)) \longrightarrow 0, \quad A \cong_X \mathcal{O}_X^{\oplus(r+1)}. \quad (4.1)$$

We may choose a decomposition

$$A = L_0 \oplus_X \cdots \oplus_X L_r, \quad L_i \cong_X \mathcal{O}_X,$$

and corresponding stably injective sections

$$t_i : \mathcal{O}_X \xrightarrow{\sim} L_i \hookrightarrow V.$$

For every point $x \in |W_{n,d}|$, at least one t_i is transverse to the kernel at x . Equivalently,

$$|W_{n,d}| = \bigcup_{i=0}^r |U_{t_i}|.$$

More precisely, if $\xi : S_x = \mathrm{Spa}(K, K^+) \rightarrow W_{n,d}$ represents x , where K is any perfectoid affinoid field, then ξ factors through at least one U_{t_i} .

Proof. The extension theorem of Birkbeck–Feng–Hansen–Hong–Li–Wang–Ye [1, Theorem 1.1.2] applies to

$$F_1 = \mathcal{O}_X^{\oplus(r+1)}, \quad F_2 = \mathcal{O}_X(1/(d-1)), \quad E = V = \mathcal{O}_X(1/n).$$

The ranks and degrees agree, we have $0 < 1/(d-1)$, and the straight Harder–Narasimhan polygon of V lies below that of $F_1 \oplus F_2$. Hence the short exact sequence (4.1) exists.

The inclusion $A \hookrightarrow V$ has vector-bundle cokernel, and each inclusion $L_i \hookrightarrow A$ is split. The local splitting argument from Lemma 2.2 therefore shows that every composite

$$t_i : \mathcal{O}_X \cong L_i \hookrightarrow A \hookrightarrow V$$

is stably injective.

We now fix a point $x \in |W_{n,d}|$ and a representative

$$\xi : S_x = \mathrm{Spa}(K, K^+) \longrightarrow W_{n,d}.$$

No algebraic closedness assumption on K is needed. We write

$$K_x \hookrightarrow V_x, \quad Q_x = V_x/K_x$$

for the corresponding kernel and quotient on X_{S_x} . The kernel K_x has rank r and degree 0. Pullback of (4.1) to X_{S_x} remains exact.

Suppose, for contradiction, that every composite

$$L_{i,x} \longrightarrow V_x \longrightarrow Q_x$$

were zero. Then the whole map $A_x \rightarrow Q_x$ would vanish, so the injection $A_x \hookrightarrow V_x$ would factor through K_x . This would produce an injection from a rank- $(r+1)$ vector bundle into the rank- r vector bundle K_x , which is impossible. Hence there is an index i for which $L_{i,x} \rightarrow Q_x$ is nonzero.

For a point S_x of arbitrary rank, restriction to $X_{\mathrm{Spa}(K, K^\circ)}$ is an equivalence on coherent sheaves by [3, §A.2]. We may therefore check injectivity on this integral rank-one curve. There a nonzero map from a line bundle to a vector bundle is injective. It follows that

$$E_x := K_x \oplus L_{i,x} \longrightarrow V_x$$

is injective.

We still have to upgrade ordinary injectivity to stable injectivity. The bundle E_x has rank $r+1$ and degree 0, whereas V_x is semistable of slope $1/n$. Since $d > 1$, we have $r+1 < n$ and therefore

$$\frac{\deg(E_x) + 1}{\mathrm{rk}(E_x)} = \frac{1}{r+1} > \frac{1}{n} = \mu(V_x).$$

Lemma A.2.1 of [3], which is stated for arbitrary points, now shows that E_x is saturated in V_x . Hence V_x/E_x is a vector bundle. The local splitting argument from Lemma 2.2 shows that $E_x \rightarrow V_x$ is stably injective. Consequently the original map ξ factors through U_{t_i} .

Since x was arbitrary, the open subsets $|U_{t_i}|$ cover $|W_{n,d}|$. $\square$

5 Proof of the main theorem

Proof of Theorem 1.2. We argue simultaneously for all $n > d$ by induction on d . For $d = 1$, Lemma 2.2 identifies $W_{n,1}$ with the functor of stably injective subbundles

$$K \hookrightarrow \mathcal{O}_{X_S}(1/n), \quad K \cong_{\text{geom}/S} \mathcal{O}^{\oplus(n-1)}.$$

Johansson–Ludwig–Hansen identify this with the quotient of the infinite-level Lubin–Tate space by the parabolic of block type $(n-1, 1)$ [3, Theorem A.1.1], and [3, Theorem A and Proposition 4.2.7] represent that quotient by a perfectoid space.

Let $d > 1$ and assume that $W_{n-1,d-1}$ is perfectoid. By Proposition 2.3, $W_{n,d}$ is a diamond. Given $x \in |W_{n,d}|$, Lemma 4.1 gives i with $x \in |U_{t_i}|$. The subdiamond U_{t_i} is open by Lemma 2.4 and perfectoid by Corollary 3.4. Hence every point has an open perfectoid neighbourhood, so Lemma 2.1 shows that $W_{n,d}$ is perfectoid. After $d-1$ steps the induction reaches the case $d = 1$. $\square$

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